

Spectral Graph Theory

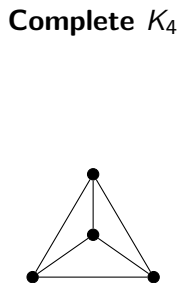
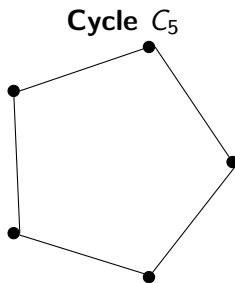
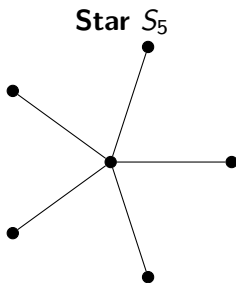
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What is a graph?

Definition: A (simple) graph is a pair $G = (V, E)$, where:

- $V = \{v_1, v_2, \dots, v_n\}$ is a finite set of **vertices**,
- $E \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is a set of unordered pairs of distinct vertices called **edges**.



The Adjacency Matrix

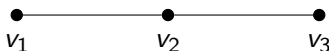
Definition: Let $G = (V, E)$ be a graph with n vertices, where $V = \{v_1, v_2, \dots, v_n\}$. The **adjacency matrix** $A = (a_{ij})$ of G is the $n \times n$ matrix defined by:

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \in E, \\ 0 & \text{otherwise.} \end{cases}$$

Example: For the graph G with $V = \{v_1, v_2, v_3\}$ and

$E = \{\{v_1, v_2\}, \{v_2, v_3\}\}$:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$



Note: We will be working with undirected graphs so A is symmetric.

Degree of a Vertex

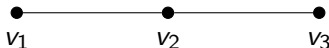
Definition: In a graph $G = (V, E)$, the **degree** of a vertex $v \in V$, denoted $\deg(v)$, is the number of edges incident to v .

Mathematically: If $A = (a_{ij})$ is the adjacency matrix of G , then:

$$\deg(v_i) = \sum_{j=1}^n a_{ij}$$

Example:

- Let G be the graph with $V = \{v_1, v_2, v_3\}$ and $E = \{\{v_1, v_2\}, \{v_2, v_3\}\}$.
- Then $\deg(v_1) = 1$, $\deg(v_2) = 2$, $\deg(v_3) = 1$.



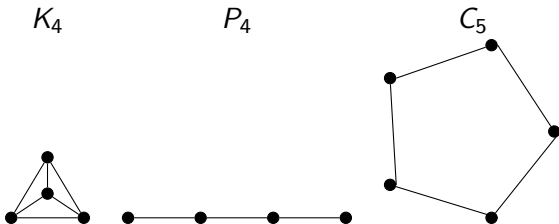
Independence Number

Definition: The **independence number** $\alpha(G)$ of a graph $G = (V, E)$ is the size of the largest subset $S \subseteq V$ such that no two vertices in S are adjacent:

$$\forall u, v \in S, \{u, v\} \notin E$$

Examples:

- In a complete graph K_n , $\alpha(K_n) = 1$
- In a path graph P_4 , $\alpha(P_4) = 2$
- In a 5-cycle C_5 , $\alpha(C_5) = 2$



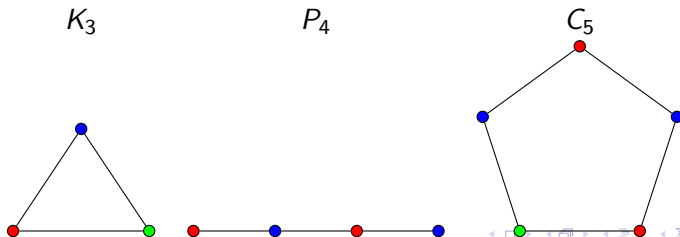
Chromatic Number

Definition: The **chromatic number** $\chi(G)$ of a graph $G = (V, E)$ is the minimum number of colors needed to assign to the vertices of G such that no two adjacent vertices share the same color:

$$\forall \{u, v\} \in E, \text{color}(u) \neq \text{color}(v)$$

Examples:

- For a complete graph K_n , $\chi(K_n) = n$
- For a bipartite graph (e.g., a path P_n), $\chi = 2$
- For a 5-cycle C_5 , $\chi(C_5) = 3$ (odd cycles require 3)



What Does NP-Hard Mean?

P vs NP:

- **P:** Problems that can be solved in polynomial time.
- **NP:** Problems for which a solution can be *verified* in polynomial time.

NP-Complete:

- Decision problems in NP that are as hard as any other problem in NP.
- If any NP-complete problem is in P, then **P = NP**.

NP-Hard:

- Problems at least as hard as NP-complete problems.
- May not be in NP (e.g., optimization problems, problems with no yes/no answer).
- No known polynomial-time solution.

Examples:

- **NP-complete:** 3-Coloring, Independent Set (decision form)
- **NP-hard:** Computing $\alpha(G)$ or $\chi(G)$ exactly

NP-Hardness of Graph Parameters

Key Idea: Many natural graph problems are computationally difficult — specifically, they are **NP-hard**, meaning no known polynomial-time algorithms exist to solve them in general.

Independence Number:

- Problem: Given a graph G and integer k , is there an independent set of size at least k ?
- This is known as the **Maximum Independent Set Problem**.
- It is NP-complete to decide, and NP-hard to compute $\alpha(G)$.

Chromatic Number:

- Problem: Given a graph G and integer k , can G be colored with k colors?
- This is the classical **Graph Coloring Problem**.
- It is NP-complete for $k \geq 3$, and NP-hard to compute $\chi(G)$.

Inapproximability:

- Approximating either $\alpha(G)$ or $\chi(G)$ within a factor of $n^{1-\epsilon}$ is

Eigenvalues of a graph

The definition is nearly identical to the definition in standard linear algebra that one might expect!

Definition: Let $A(G) = (a_{ij})$ be the adjacency matrix of a graph G . The eigenvalues of the graph G are the eigenvalues of the adjacency matrix of G . That is, the values of $\lambda \in \mathbb{R}$ such that

$$A\mathbf{x} = \lambda\mathbf{x}$$

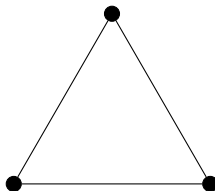
where $\mathbf{x}$ is some nonzero vector in $\mathbb{R}^n$.

Definition: We call the set of eigenvalues of a graph G the **spectrum** of G .

Simple example of K_3

The eigenvalues of a simple graph like K_3 are rather simple to find!

The graph of K_3 is as follows:



The adjacency matrix of K_3 is

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Using standard linear algebra computations on the adjacency matrix, the eigenvalues of K_3 are 2 (multiplicity of 1) and -1 (multiplicity of 2).

K_3 example cont'd

The associated eigenvectors of the eigenvalues are $(1,1,1)$ for $\lambda = 2$, $(-1,1,0)$ and $(-1,0,1)$ for $\lambda = -1$.

A few observations:

- Notice that the largest eigenvalue, $\lambda = 2$ is precisely the degree of every vertex! Does it always hold true?
- There is also some sort of a "flipping nature". Notice, for $\mathbf{v} = (-1, 1, 0)$, if we apply the adjacency matrix to it, so

$$A\mathbf{v} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = -\mathbf{v}!$$

This is due to the definition of the eigenvalue (since $\lambda = -1$), and it flips the direction of $\mathbf{v}$.

Perron-Frobenius

Theorem (Perron-Frobenius for the adjacency matrix): Let G be a connected graph. Let $A(G)$ be the adjacency matrix of G . Then, the largest eigenvalue λ_1 of $A(G)$ is simple, with a corresponding eigenvector whose entries are all positive. Moreover, for all eigenvalues λ of $A(G)$, $|\lambda| \leq \lambda_1$.

Connectedness of a graph

So, why eigenvalues? How does it help measure connectivity of a graph?

Proposition: Let Δ be the maximum degree of a graph G . Let $\bar{d}$ be the average degree. Then

$$\max\{\sqrt{\Delta}, \bar{d}\} \leq \lambda_1 \leq \Delta$$

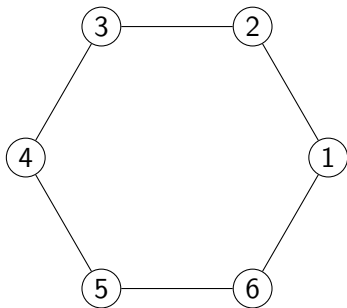
What happens when λ_1 is small? When it is large?

Connectedness of a graph

We now know that λ_1 is a good measure of connectivity. It turns out λ_2 is also a good measure of connectivity in regular graphs!

Definition: A graph is **d -regular** if and only if the degree of each of the vertices have degree d .

Example: Here is a 2-regular graph.



Regular graphs and connectedness

It turns out in a d -regular graph, $\lambda_1 = d$.

Proposition: Let G be a d -regular graph. Then $\lambda_1 = d$ and its multiplicity is the number of components. Every eigenvector belonging to d is constant on each component.

This motivates the following theorem...

Bounds on edges in a d -regular graph

- Edges are a wonderful tool to measure connectivity. A vertex having a lot of edges means that it has a lot of "neighbours".
- Could we extend such notion to sets of vertices to measure global connectedness rather than local connectedness? With the help of λ_2 , yes!

Theorem: Let G be a d -regular graph on n vertices with eigenvalues $d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. Let $S, T \subseteq V(G)$ such that $S \cup T = V(G)$ and $S \cap T = \emptyset$. Then

$$(d - \lambda_2) \frac{|S||T|}{n} \leq e(S, T) \leq (d - \lambda_n) \frac{|S||T|}{n}$$

Definition: We call $d - \lambda_2$ the **spectral gap** of a graph G .

What happens if there is a large spectral gap?

Definition of Laplacian Matrix

To get deeper insight into a graph's connectivity, we look at a Laplacian matrix.

Definition: Let G be a graph without loops. The Laplacian matrix $L(G)$ of the graph G is defined as follows: the diagonal elements are $L(G)_{ii} = d_i$, the degrees of the vertices, and if $i \neq j$, then $L(G)_{ij} = -a_{ij}$, where a_{ij} is the number of edges between vertices i and j . In simpler terms, $L(G) = D - A$, where A is the adjacency matrix and $D = \text{diag}(d_1, \dots, d_n)$.

Motivation for Laplacian Matrix

- The Laplacian matrix is symmetric and positive semi-definite, making it easier to apply calculus and linear algebra tools.
- Used for energy minimization of functions over a graph.
- The Laplacian introduces a new notion of connectivity when compared to the Adjacency matrix.

Eigenvalues of Laplacian Matrix

A key property of the Laplacian that makes it particularly useful for analysis is that its eigenvalues are always non-negative, which not only simplifies computations but also provides valuable insights into the graph's structure, such as the number of connected components.

Theorem: Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ be the eigenvalues of $L(G)$. Then $\lambda_n = 0$. 0 is an eigenvalue of G and all eigenvalues are non-negative.

Theorem: The number of components in G is given by the multiplicity of the 0 eigenvalue of $L(G)$

Algebraic Connectivity

- The second smallest eigenvalue of the Laplacian of a graph G is called the algebraic connectivity, denoted by $a(G)$.
- $a(G) = 0$ iff G is not connected.
- $a(G) = n$ iff G is complete.

Laplacian of K_3

Given K_3 , a complete graph on 3 vertices, the adjacency matrix of K_3 is

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

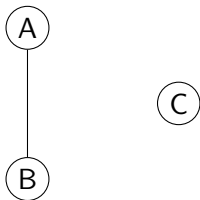
Since each vertex has degree 2, $D = \text{diag}(2, 2, 2)$ for K_3 , giving us the following Laplacian matrix:

$$\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

Computing the eigenvalues, we see that $\lambda = (0, 3, 3)$, highlighting the fact that K_3 is a connected graph.

Laplacian of a Disconnected Graph

Consider P_1 , a path graph on two vertices, and P_2 , a graph on an isolated node.



The adjacency matrix of $P_1 \cup P_2$ is:

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Laplacian of a Disconnected Graph

Since vertex A and B have degree 1 and vertex C has degree 0, $D = \text{diag}(1, 1, 0)$ for $P_1 \cup P_2$, giving us the following Laplacian matrix:

$$\begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Computing the eigenvalues, we see that $\lambda = (0, 0, 2)$. Given this information, we can say that the graph has two components since the multiplicity of the 0 eigenvalue of the Laplacian matrix is 2.

Spectral Graph Drawings

We want to visualize a graph. Suppose $|V| = n$, we want to map each $v \in V$ to a point in $\mathbb{R}^d$ where $d \ll n$. For example, $d = 2$.

Definition: A graph drawing is a function $\rho : V \rightarrow \mathbb{R}^d$.

Definition: The matrix of a graph drawing ρ is the $m \times n$ matrix whose i -th row consists of the row vector $\rho(v_i)$ corresponding to the points in $\mathbb{R}^d$.

Energy Minimization

Motivation: We can imagine building a physical model of a graph G by connecting adjacent vertices in $\mathbb{R}^d$ with identical springs. It would be natural to consider a representation to be better if it requires the springs to be less extended.

Definition: The energy of a drawing ρ is

$$\mathcal{E}(\rho) = \sum_{\{v_i, v_j\} \in E} \|\rho(v_i) - \rho(v_j)\|^2$$

A “good drawing” is one that minimizes the energy function $\mathcal{E}$.

Algorithm for Laplacian Graph Drawing

To draw a graph nicely in 2D:

- 1 Compute the eigenvectors of the second and third smallest eigenvalue of the Laplacian.
- 2 Place them as column vectors in a matrix.
- 3 Place each vertex at the coordinate given by the rows.

The resulting graph usually produces a visually pleasing layout, while minimizing “spring energy”.

Example: Spectral Drawing of a 4-Node Graph

Graph:

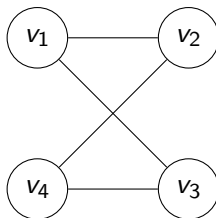
$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}, \quad L = \begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & 0 & -1 \\ -1 & 0 & 2 & -1 \\ 0 & -1 & -1 & 2 \end{bmatrix}$$

Eigenvectors Matrix for $\lambda_2 = \lambda_3 = 2$:

$$R = [u_2 \ u_3] = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}, \quad \text{So } \begin{cases} v_1 \rightarrow (0.5, 0.5) \\ v_2 \rightarrow (0.5, -0.5) \\ v_3 \rightarrow (-0.5, -0.5) \\ v_4 \rightarrow (-0.5, 0.5) \end{cases}$$

Example: Spectral Drawing of a 4-Node Graph

Resulting 2D Drawing:



More on spectral graph drawing: Demo